ELEC-ACC-01: Arista Access Node Routing Protocol Flap Resolution SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4757

Description

The Electronic City Hub Access Node (ELEC-ACC-01) running Arista's EOS operating system is experiencing a Routing Protocol Flap, characterized by BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This phenomenon occurs when the Access Node's role and vendor equipment configuration inadvertently exacerbate BGP's inherent instability, particularly during high-speed network reconfigurations. Arista's EOS, being a multi-protocol routing platform, is sensitive to the nuances of BGP's route reflection and iBGP confederations, which can lead to oscillations in the BGP routing table. Furthermore, the Access Node's involvement in OSPF and ISIS routing protocols introduces additional complexities, as these protocols are designed to adapt to changing network topologies, potentially inducing adjacency loss and LSP flooding. In this context, the ALARM_4757, indicating a Warning severity, is triggered when the BGP route churn and OSPF adjacency loss reach a critical threshold, necessitating immediate attention to prevent network instability and potential service disruptions.

Pre-requisite Checks

- check: Verify redundant link status for ELEC-ACC-01 at Electronic City Hub.
- check: Confirm remote hands' readiness for dispatch to Electronic City Hub.
- check: Ensure network monitoring tools are alerting on the ALARM_4757 for ELEC-ACC-01.

Potential Root Causes

Cause	Probability
BGP route reflection instability.	High
Incorrect iBGP confederation configuration.	Medium
OSPF adjacency loss due to misconfigured priority settings.	Low
ISIS LSP flooding caused by suboptimal routing configuration.	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 100, main routing table version 100, Description: This command verifies the BGP table version and main routing table version, ensuring synchronization between the two.

Command: show ip ospf database, Expected Output: OSPF database has 100 routes, Description: This command displays the OSPF database, helping identify potential adjacency loss or routing inconsistencies.
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Command: show isis database, Expected Output: ISIS database has 100 LSPs, Description: This command shows the ISIS database, aiding in the identification of potential LSP flooding or routing anomalies.
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Resolution Steps

Step 1: Step: Stabilize BGP route reflection, Command: configure terminal router bgp 65000 neighbor 10.10.10.1 remote-as 65000 neighbor 10.10.10.1 update-source Loopback0, Description: Configure BGP neighbor relationships with a stable update source, ensuring consistent routing information exchange.

Step 2: Step: Review and correct iBGP confederation configuration, Command: configure terminal router bgp 65000 bgp confederation identifier 65000 bgp confederation peers 65001, Description: Verify and correct the iBGP confederation configuration to prevent unnecessary route reflection and instability.

Step 3: Step: Adjust OSPF priority settings, Command: configure terminal router ospf 100 interface Ethernet1/1 ospf priority 150, Description: Adjust OSPF priority settings to prevent adjacency loss and ensure stable routing.

Step 4: Step: Reconfigure ISIS routing, Command: configure terminal router isis net 49.0001.0100.0000.1000.00, Description: Reconfigure ISIS routing to prevent LSP flooding and ensure optimal routing.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 100, main routing table version 100, Description: Verify BGP table consistency and synchronization.

Command: show ip ospf database, Expected Output: OSPF database has 100 routes, Description: Verify OSPF database consistency and routing stability.

Command: show isis database, Expected Output: ISIS database has 100 LSPs, Description: Verify ISIS database consistency and routing stability.
```

Roll-back Procedures

- Step: Undo BGP route reflection stabilization, Command: configure terminal router bgp 65000 no neighbor 10.10.10.1 remote-as 65000, Description: Undo BGP neighbor relationships and return to previous configuration.

- Step: Revert iBGP confederation configuration, Command: configure terminal router bgp 65000 no bgp confederation identifier 65000, Description: Revert iBGP confederation configuration to previous state.

- Step: Revert OSPF priority settings, Command: configure terminal router ospf 100 no interface Ethernet1/1 ospf priority 150, Description: Revert OSPF priority settings to previous state.

- Step: Revert ISIS routing configuration, Command: configure terminal router isis no net 49.0001.0100.0000.1000.00, Description: Revert ISIS routing configuration to previous state.